The Local Economic Impact of Data Centers

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Abstract

A hyperscale data center costs over a billion dollars but employs only a few dozen people. We ask whether this investment generates local agglomeration spillovers. Within-site ring comparisons show a large construction footprint in nighttime lights that fades within five kilometers. Advertised salaries and new-firm registrations do not rise; hiring and supplier estimates are also near zero but less precise. County data show direct data-processing entry, not a downstream cluster. Compute firms locate near data centers, but most arrive first. Rent estimates are positive but uncertain. Digital capital can transform a site without producing a broad local economic response.

Keywords: data centers; agglomeration; economic geography; place-based investment.

JEL Codes: R11, R23, O33, H71.

1 Introduction

The physical infrastructure of digital services has become a major component of US capital investment. Data centers consumed roughly four percent of US electricity in 2023, and the rise of artificial intelligence has accelerated new construction [Shehabi et al., 2024, International Energy Agency, 2025]. A hyperscale campus can cost more than a billion dollars and draw hundreds of megawatts of continuous power, yet employs only a few dozen people on site [Alvarez et al., 2026]. At least thirty-five states now offer tax incentives aimed specifically at data centers, with cumulative subsidies approaching twenty billion dollars.[1] Those subsidies are justified in part by the promise of local economic gains. This paper asks whether the gains materialize.

The natural benchmark is the canonical result on large plant openings. Greenstone et al. [2010] compare counties that won a large new plant to the runner-up counties that lost it and find that the productivity of incumbent establishments rises in the winning counties. The gain operates through three agglomeration channels: a deeper shared labor pool, denser supplier linkages, and knowledge spillovers among nearby firms. This study provides influential evidence for place-based investment policy. At the same time, subsequent work finds that winner-versus-loser estimates are sensitive to the comparison counties [Patrick, 2016], and that the average large plant produces little evidence of productivity spillovers [Patrick and Partridge, 2022]. The spillover evidence is therefore contested even for manufacturing plants, the historically strongest case for agglomeration.

A data center separates the location of capital from many of the workers, suppliers, and customers that use it. This makes the sector a useful test of the micro-foundations of agglomeration. Labor pooling, input sharing, and knowledge spillovers can all make proximity productive [Duranton and Puga, 2004, Ellison et al., 2010]. A data center, however, employs dozens of people rather than thousands. Its main inputs include electricity, network connectivity, construction services, and specialized equipment, much of which can be purchased from regional or national suppliers. Its output is delivered over the network to customers around the world. Capital scale alone may therefore be insufficient to generate a local multiplier.

Operators screen sites on cheap power, grid capacity, land cost, and fiber, so a county that hosts a data center differs from one that does not in ways that also shape the local economy. We do not find a credible instrument for this selection. The predetermined features that predict siting are correlated with local industrial conditions, while climate and historical human capital do not predict hyperscale locations (Section 3).

We use three designs for different outcomes. The within-site design compares an inner ring around each facility with an outer ring around the same site and is our main spatial design for outcomes with a flat pre-period. We date construction from satellite-detected land clearing and restrict the capital analysis to sites without an earlier data center within two kilometers. We also assemble eighty-four announced hyperscale-sized projects that were not built, of which fifty-two have satellite coverage. These projects crossed an observable announcement and siting threshold,

[1] Authors' tally from the Good Jobs First Subsidy Tracker, which records state and local data-center subsidy awards and is one of the sources for our facility registry (Section 2).

but they are not literal runner-up locations. They are newer, farther from existing data-center clusters, and less likely to have a named hyperscale sponsor than the built sites. We therefore use them as an announced near-miss benchmark, not as an assumption that the groups are balanced. County panels provide a third diagnostic for industry outcomes, although suppression and pre-opening trends limit what these data establish.

We first establish that the capital footprint is large and local. At isolated sites, nighttime lights rise by 38 percent within the first kilometer when construction begins and decay to near zero by five kilometers. The announced near-miss comparison produces a larger construction-year estimate and the same timing. Daytime imagery confirms that vegetation clears when the lights appear.

The evidence for agglomeration depends on the outcome. Advertised salaries and new-firm registrations do not rise, and their confidence intervals exclude increases above 5.2 and 1.6 percent. A national county analysis of business applications gives a similar result. Hiring and supplier-posting estimates are also centered near zero, but their confidence intervals admit increases of roughly thirty percent. Workplace employment rises before opening rather than breaking at the event. At the county level, data-processing establishments rise by 27 percent, consistent with the direct entry of the facility. Apparent establishment growth in broader information and professional services is also present before opening.

The geography of compute reinforces this distinction. Compute-intensive firms locate closer to data centers than other firms, but 69 percent of those within 25 kilometers predate the nearest facility. The median metropolitan data center has thousands of compute firms nearby, while the median exurban data center has 41. The two populations share metropolitan infrastructure, skilled labor, and access to the power and fiber backbone. Their geographic overlap does not imply that the data center created the cluster.

The non-labor evidence is more suggestive. Synthetic-control rent estimates are two to five percent, but the announced near-miss estimate is imprecise and the synthetic comparison does not support formal placebo inference. Multifamily permitting and net migration do not rise. School property-tax revenue and county highway spending have positive point estimates, but neither survives adjustment across the fiscal outcomes. Foot traffic and commercial spending show no robust change. Residential electricity prices, reported in the appendix, do not rise for the cloud-era cohorts in our sample.

Concurrent county-level studies find positive effects of data-center entry or capacity on employment, establishments, income, construction, and house prices [Bahar and Wright, 2026, Alvarez et al., 2026, Yue and Zeng, 2026]. Our question is different. County estimates combine the direct facility, construction, and growth already under way in selected locations. We ask whether construction or opening produces a discrete response among nearby workers, suppliers, and firms, and we trace where compute users were located first. The results reconcile a large direct investment with limited evidence of a new local cluster. This distinction matters as place-based investment shifts from labor-intensive plants toward labor-light digital capital.

2 Data

Data center registry. We build a master registry of US data centers from several sources. The starting point is the Open Source Data Center Atlas (Pacific Northwest National Laboratory's IM3 project), which identifies 1,242 facilities from OpenStreetMap footprints tagged as data centers, with coordinates and, where available, square footage and operator. We add operational-date information from operator press releases, state economic-development announcements, the Good Jobs First subsidy tracker, and trade press. We also conduct targeted searches of state announcements, specialty wholesale operators, AI-wave operators, and real-estate investment trust filings. The final registry contains 1,494 facilities. The geocoded analysis sample contains 1,118 facilities, including 341 classified as hyperscale. Of the hyperscale facilities, 315 have high- or medium-confidence analysis dates inside the 2012–2024 VIIRS window. We classify facilities as hyperscale based on operator identity (Amazon, Microsoft, Google, Meta, Apple, Oracle, plus hand-curated large specialty and AI-wave campuses). We do not impose a megawatt threshold, since capacity is not consistently observed. We also assemble an announced near-miss cohort for the built-versus-near-miss benchmark. It contains eighty-four announced projects that were not built, of which fifty-two have the coordinates and satellite coverage to enter the nighttime-lights design. Section 3 reports the cohort's sponsor composition and shows that the capital result is robust to restricting the control group to projects with a named hyperscale tenant.

Table 1: Sample construction

Sample restriction	N facilities
Master registry (all sources, all years)	1,494
geocoded and dated analysis backbone	1,118
in 2012–2024 (VIIRS window)	1,110
with clean-confidence analysis date	1,088
Hyperscale-tier facilities	341
in VIIRS window 2012–2024	334
with clean-confidence analysis date	315
with satellite-detected construction date	87
Isolated built sites in capital design	**43**
Isolated announced near-miss sites	**49**

Notes: Sample is constructed from a master registry of 1,494 US data centers compiled from the IM3 OpenStreetMap base, operator press releases, state economic-development announcements, Good Jobs First subsidy tracker entries, and industry press. Hyperscale facilities include the six major cloud operators and hand-classified large specialty and artificial-intelligence campuses. The date audit (`cf_395`) stores announcement, groundbreaking, construction, planned-completion, and operational events separately. Clean dates are high or medium confidence; mechanically projected openings, planned-completion dates without verified operation, and unresolved source conflicts are low confidence. The capital design further requires a satellite-detected construction date and no earlier data center within two kilometers.

Nighttime lights. We extract monthly mean radiance from the Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band Monthly Composite for April 2012 through December 2024, in concentric rings of 0–1, 1–2, 2–5, 5–10, 10–25, and 25–50 kilometers around each facility, aggregated to annual means. We classify each facility as isolated if no other data center within two kilometers opened earlier, and use the isolated, construction-dated sample for the headline capital-footprint estimate.

Local labor, housing, and spending panels. We draw four panels from Dewey Data. LinkUp Job Records [LinkUp, 2025] measures job postings by employer, geocoded location, ring, and quarter. RentHub Rental Data [RentHub, 2022] measures monthly median list prices by ring. We measure consumer spending with SafeGraph Spend Patterns [SafeGraph, 2022] and foot traffic with Advan Weekly Patterns [Advan Research, 2025]. Each enters the within-site ring design dated to operations, with state-by-month fixed effects.

County employment. For the county-level analysis, we use industry employment and establishment counts from the Quarterly Census of Employment and Wages. We retain the Bureau of Labor Statistics disclosure flag, which is essential because the public files record suppressed employment as zero even though the value is not observed. We focus on data processing (NAICS 518), telecommunications (517), web and information services (519), IT consulting (5415), and professional, scientific, and technical services (541). We end the 519 series in 2021 because the 2022 NAICS revision moved major components of that sector to 513 and 516. Section 5.5 reports both the disclosure rates and the pre-trend diagnostics that limit causal interpretation of these county outcomes.

Instrument-search inputs. The instrument search draws on predetermined county features, namely 345 kV transmission-substation density, the InterTubes long-haul fiber network [Durairajan et al., 2015], the 1980 county share of urban college population (NHGIS), station-level climate normals (CustomWeather), and the EIA-860 generation fleet (Appendix B).

2.1 Dating construction and operations

We preserve announcement, groundbreaking, construction, planned-completion, and operational dates as separate events. We assign high confidence to operational dates reported by operators or public agencies. We assign medium confidence to less detailed sources when the year agrees with an independent registry or satellite evidence. Announcement-based projections, planned-completion dates, and unresolved source conflicts remain low confidence and are excluded from analyses restricted to reliable dates. This classification leaves 315 high- or medium-confidence hyperscale dates and 26 low-confidence dates. For a smaller sample, we recover construction timing from daytime satellite imagery. A hyperscale build begins with land clearing. In Landsat and Sentinel-2 imagery, clearing appears as a sustained drop in the parcel's vegetation index combined with a rise in its built-up-surface index. Vegetation cover at the parcel is flat until the construction year and falls sharply afterward. As an independent check, we match Federal Aviation Administration obstruction-evaluation filings to our projects. The imagery sample contains 43 hyperscale facilities, while the independent FAA matches contain 64 facilities within 500 meters and 119 within one kilometer (Section 3.5). These independent dates validate the satellite construction timing used in the capital design.

3 Research Design

Data centers do not locate at random. Operators screen on cheap power, available grid capacity, large parcels of low-cost land, fiber proximity, and tax policy, so a county that hosts a data center differs from one that does not in ways that also move the local economy. A cross-county comparison would confound the data center with the conditions that attracted it. We document this selection directly. We test a range of predetermined siting predictors as potential instruments and reject each for reasons reported in Appendix B. The predictors that strongly predict siting—power, industrial land, and fiber—are themselves correlated with local industrial conditions. Climate and historical human capital are less likely to violate the exclusion restriction, but they do not predict hyperscale siting.

We therefore use three designs that answer different parts of the question. The within-site ring comparison provides the main spatial estimate for outcomes with a flat pre-period. It compares change close to a facility with change farther from the same site, absorbing time-invariant siting differences. Announced projects that were not built provide an external near-miss benchmark. Unlike the runner-up counties in Greenstone et al. [2010], these projects are not known finalists in the same site-selection process, so we do not assume that they are balanced with built sites. County panels provide an aggregate industry diagnostic, but suppression and pre-opening trends prevent us from treating most of these estimates as clean downstream employment effects.

3.1 Announced near-miss facilities

An announced and sited facility that was never built provides a benchmark for a built facility. Both groups crossed an observable siting threshold, but the comparison does not eliminate selection between projects that proceed and projects that do not. We assemble eighty-four announced-then-cancelled hyperscale-sized projects from trade-press coverage, regulatory filings, and operator announcements, of which fifty-two enter the nighttime-lights design, and estimate, for each outcome,

$$y_{i,r,t} = \sum_k \left[\beta_k \, \mathbf{1}[t - t_i^c = k] \times \text{Near}_{i,r} \times \text{Built}_i + \delta_k \, \mathbf{1}[t - t_i^c = k] \times \text{Near}_{i,r}\right] + \alpha_{i,r} + \lambda_t + u_{i,r,t}, \quad (1)$$

where Built_i marks facilities actually built and t_i^c is the satellite-detected construction-start date for built sites and the announcement date for cancelled sites. The cancelled projects do not have an observed or consistently reported scheduled start, so their timing is not symmetric with the built group. The path of β_k traces the change around built facilities relative to the change around cancelled sites. For the capital footprint, the comparison is especially sharp because a parcel that is never built does not light up, whatever the reason for cancellation.

The announced near misses vary in sponsor commitment. Of the eighty-four candidate projects we track, fourteen had a named hyperscale operator publicly attached at cancellation, nineteen were sponsored by established data-center operators without a named tenant, and the remaining fifty-one, about sixty percent, were speculative developments by real-estate sponsors. Speculative developments have no signed tenant or confirmed grid allocation and are less informative about the counterfactual for a committed project. For the capital footprint, however, a parcel that is never built does not light up whatever the sponsor. The construction-year contrast also does not depend on the speculative majority. Restricting the controls to the nine isolated projects with a named hyperscale operator leaves the groundbreaking-year jump essentially unchanged, at seventy percent versus sixty-eight percent in the full isolated sample. This subset is too small to support estimates at longer horizons. The full cancelled sample is also limited, and most cancellations followed local opposition, so cancelled sites may sit in somewhat less development-friendly places.[2] This is in part why we treat this design as a benchmark and pair it with the ring comparison below. In addition, built sites are dated by satellite-detected land clearing, while near-miss sites have less reliable scheduled dates from filings and trade press. However, the independent regulatory-filing dates described in Section 3.5 validate the built-site dates as well.

Table 2 makes the limits of the comparison explicit for the exact capital-footprint sample. Built and near-miss sites have similar local population and high-voltage access. They differ substantially in event year, baseline radiance, proximity to an earlier data center, region, and sponsor commitment. Facility-by-ring fixed effects absorb the level differences, and the pre-event path tests whether the inner-to-outer change was already diverging. The table nevertheless rules out interpreting the near misses as conventionally balanced runner-up sites.

[2] Because most near-miss projects were announced late in the sample, the near-miss group contributes few post-event observations. Coverage at event years zero through four is 48, 21, 5, 3, and 1 sites, so the built-versus-near-miss path at longer horizons is identified primarily from changes among built sites relative to the common pre-event trend rather than from post-period movement among near-miss sites. Section 4 quotes magnitudes only through the year after groundbreaking for this reason.

Table 2: Built and announced near-miss sites in the capital-footprint sample

Characteristic	Built mean	Near miss mean	Standardized difference
Event year	2018.81	2024.37	-4.21
Log population within 10 km	10.54	10.42	+0.06
Pre-event log radiance, 0–1 km	0.54	2.26	-1.28
Log distance to earlier data center	2.35	3.64	-1.19
Log distance to ≥230 kV substation	1.87	1.90	-0.04
Log distance to long-haul fiber	2.15	1.87	+0.29
Share Northeast	0.00	0.06	-0.36
Share Midwest	0.23	0.41	-0.38
Share South	0.35	0.43	-0.16
Share West	0.42	0.10	+0.76
Share with named hyperscale sponsor	1.00	0.18	+2.95

Notes: Exact isolated-site sample used at event year zero in Figure 1: 43 built sites and 49 announced near-miss sites. Built sites are dated by satellite-detected construction start; near-miss sites are dated by announcement. Distances are in kilometers before applying $\log(1 + x)$. Population is based on 2020 Census tract population centroids. High-voltage access uses substations rated at least 230 kV; long-haul fiber uses the InterTubes conduit map. The table is a diagnostic, not a claim that the groups are balanced.

3.2 Within-site rings

The near-miss comparison provides an external benchmark but is too small and imbalanced to carry every outcome. Our main spatial design compares an inner ring around a facility to the same facility's twenty-five to fifty kilometer outer ring,

$$\log(1 + \mathrm{NTL}_{i,r,t}) = \sum_k \beta_k \, \mathbf{1}[t - t_i^c = k] \times \mathrm{Near}_{i,r} + \alpha_{i,r} + \gamma_{r,t} + \varepsilon_{i,r,t}, \tag{2}$$

where i indexes the facility, r the ring, t the year, t_i^c the facility's construction-start year, $\mathrm{Near}_{i,r}$ marks the inner ring, $\alpha_{i,r}$ is a facility-by-ring fixed effect, and $\gamma_{r,t}$ is a ring-by-year fixed effect. The facility-by-ring fixed effect absorbs any time-invariant difference between a site's inner and outer ring, including the operator's siting preferences, the local industrial structure, and grid and fiber access. Identification comes from the change in the inner ring relative to the outer ring of the same site.

Two features of the design address a pre-trend present in a simpler version of this specification. First, we date the event to construction start, not to the operational opening, to avoid confusing pre-period construction activity with operations (Section 2.1). Dated to construction, the inner-ring brightness is flat before groundbreaking and jumps at the groundbreaking date.

Second, we restrict the headline estimate to isolated sites, meaning facilities with no earlier data center within two kilometers. A new facility built beside an existing one has a brighter inner ring even before construction begins, which could generate a false pre-trend. The isolated, construction-dated event study has a flat pre-trend and then a jump of thirty-eight percent in the construction year that builds to a larger footprint over the following years. The jump survives controlling for a site-specific linear pre-trend estimated across the pre-construction years (the pre-trend is flat).

We report ring estimates only for outcomes whose pre-construction estimates are flat. The capital footprint, rents, advertised wages, and firm entry pass this test. Job-posting counts and workplace employment do not, so for those outcomes we rely on the announced-near-miss design and do not interpret the ring estimates in either direction. To make these choices transparent, Table 3 reports every outcome under every design in which it can be estimated, with the estimate, its precision, and the reason any design is excluded for that outcome.

3.3 Demand-side outcomes

We use the same within-site ring comparison in equation (2) for three demand-side outcomes: residential rents, consumer spending, and foot traffic. We date these outcomes to the operational opening rather than to the construction start. The rent, spending, and foot-traffic panels are monthly and span the 2020 to 2023 period, so we replace the ring-by-year fixed effect with a state-by-month fixed effect, which absorbs the differential work-from-home and urban-recovery shocks across metropolitan areas in those years. The announced near-miss design provides a second comparison where those data are available. Relative to near-miss sites, rents rise modestly at built sites, foot traffic shows no robust change, and commercial spending does not change in either comparison. The rent, spending, and foot-traffic estimates enter the incidence analysis in Section 6.

3.4 Employment and county-industry diagnostics

Local labor demand is harder to identify than the capital footprint, so we measure it in two ways. The first approach applies the within-site ring design of equation (2) to job postings near each facility, counting openings advertised within each ring.

The second approach is at the county level, because a data center's own employees and any firms that cluster around it need not fall inside a single ring. We use Quarterly Census of Employment and Wages employment and establishment counts by industry to estimate

$$\log(1 + \mathrm{Emp}^s_{c,t}) = \sum_k \theta^s_k \, \mathbf{1}[t - t^o_c = k] \times \mathrm{DC}_c + \mu_c + \lambda_t + u_{c,t}, \tag{3}$$

for industry s, county c, and the county's first data-center year t^o_c, with county and year fixed effects. We also estimate cohort-by-event-time effects using not-yet-treated counties as controls and state-cluster bootstrap inference. These diagnostics limit how we use the county data. Between 27 and 68 percent of the detailed employment cells are suppressed, depending on sector, and establishment counts in several sectors rise before opening. We therefore use these outcomes to show why the county evidence cannot cleanly identify either a downstream gain or a precise zero. Section 5.5 reports the diagnostics and results in full.

3.5 Credibility of the capital-footprint design

The capital-footprint estimate is robust to the usual checks on a ring design. Dropping each facility in turn leaves the estimate in a narrow band, so no single site drives it. A specification that omits the campus itself and uses the one-to-two kilometer ring as the inner comparison returns the same spatial decay. Because most hyperscale sites sit near one another, we cluster standard errors at the campus, rather than the facility, level, which widens the interval but leaves the effect statistically significant. Most important for construction timing, we re-date the facilities from Federal Aviation Administration (FAA) obstruction-evaluation filings. These regulatory timestamps are recorded years before any radiance response, and we match them to facilities by location without using satellite data. The construction-dated capital footprint is similar in sign and total magnitude under these independent dates. The FAA filing typically precedes physical construction, so this is a timing validation rather than an exact alternative groundbreaking date.

In Section 6, we conduct one additional check for the rent result. Rents rose generally over 2020 to 2023 in the kinds of places that host data centers, and a within-site comparison would attribute that growth to the facility. We therefore build a synthetic control for each facility from population-matched locations with no data center. This comparison accounts for common growth and isolates the data-center-specific rise.

Table 3 records which design carries the interpretation for each outcome. This routing matters because no single comparison is credible for all outcomes.

Table 3: Outcome estimates across research designs

Outcome (hyperscale, inner ring)	Built vs. near miss	Ring / synthetic	County evidence
Capital (nighttime lights)	**+70% ($p < 0.01$)**	**+38% ($p < 0.01$)**	—[a]
Residential rent	+4.3% ($p = 0.23$)	**+2 to +5%**[b]	—[a]
Advertised salary	−11.6% ($p = 0.54$)	**−0.9% ($p = 0.77$)**	—[a]
Job postings	**+4.0% ($p = 0.75$)**[c]	excluded[d]	—[a]
Foot traffic	**−12.0% ($p = 0.12$)**[e]	excluded[d]	—[a]
Consumer spending	**+4.6% ($p = 0.60$)**	+8.5% ($p = 0.50$)[f]	—[a]
Firm entry (SoS registrations)	—[g]	**−0.5% ($p = 0.62$)**[h]	+1.4% ($p = 0.49$)[i]
Workplace employment	—[g]	excluded[d]	inconclusive[j]

Notes: Post-event estimates for hyperscale facilities at the innermost ring each design supports (0–1 km for nighttime lights, 0–5 km otherwise). Bold marks the design used to interpret each outcome. The built-versus-near-miss capital estimate is the isolated-sample construction-year jump; the two groups are not balanced on several observable characteristics (Table 2), and magnitudes at longer horizons rest on few near-miss controls (Section 4).

[a] Not estimated at the county level; no county analog of the ring outcome.

[b] Synthetic-control headline; the within-site estimates (+6.6 to +9.4%) are an upper bound (Section 6).

[c] The estimate using updated opening dates is imprecise. Alternative count, long-difference, and construction-dated specifications likewise do not identify a precise hiring response. Minimum detectable increases remain large.

[d] Excluded because the pre-opening coefficients show an inner-versus-outer trend that the ring design cannot separate from an effect (Section 3).

[e] Falls to −3.0% ($p = 0.68$) on the balanced panel, so we do not interpret it as reliable evidence of displacement.

[f] Reported with caveats in Section 6; passes a random-donor comparison.

[g] No panel around near-miss sites exists for this outcome (business registrations and LODES cannot be reconstructed around never-built sites at ring scale).

[h] Hyperscale-only sample using updated opening dates; the pooled all-archetype estimate is +0.7% ($p = 0.24$).

[i] Census Business Formation Statistics, national county design.

[j] Data-processing establishments rise with direct entry. Detailed QCEW employment is heavily suppressed, while establishment counts in several downstream sectors rise before opening. Estimates that preserve the suppression flags and use not-yet-treated controls therefore do not identify either a downstream gain or a precise zero (Section 5.5).

4 The capital footprint

4.1 Announced near-miss benchmark

Figure 1 reports the announced near-miss benchmark from equation (1). The design compares the inner-ring (zero to one kilometer) nighttime-lights path at built hyperscale sites to announced projects that were not built. Built sites are dated to satellite-detected construction start; near-miss sites are dated to announcement because they do not have an observed construction start. We focus on *isolated* sites, which are the forty-three built facilities with no earlier data center within two kilometers of the parcel and the forty-nine near-miss sites away from operating campuses. As Table 2 shows, these groups are not balanced on several observable characteristics. The pre-construction coefficients are between eight and eleven percent below the omitted year. They are jointly indistinguishable from zero when clustered by campus ($p = 0.24$), and their linear slope is

+3.5 log points per year ($p = 0.22$). Controlling for that slope leaves a sixty-nine percent jump in the construction year ($p < 0.01$). The event-study estimate is seventy percent in the construction year and reaches roughly two hundred thirty-one percent above the near-miss benchmark a year later. The measured footprint reflects activity at the parcel, its security and parking lighting, substations and cooling equipment, and continuous operation.

Two properties of this estimate matter for how it should be read. First, the near-miss cohort is concentrated in recent announcement years, so its coverage declines with the event horizon. Forty-eight control sites identify the construction-year contrast and twenty-one the year-one contrast, but five or fewer identify the contrasts beyond that. We therefore end the figure after year one and do not use the later coefficients. Second, the footprint belongs to the campus, not to a single building. Thirty-two of the forty-three isolated built sites add a sibling building within two kilometers during the event window, so the multi-year build-up reflects additional buildings being phased in. Restricting to the eleven sites that stay single-building through year four, which are also the smallest projects, gives a footprint of about seven percent at groundbreaking, imprecisely estimated. We report the campus-level contrast because the campus is what a jurisdiction recruits. Clustering standard errors at the campus level widens the intervals by roughly sixty percent and leaves every post-groundbreaking coefficient statistically significant.

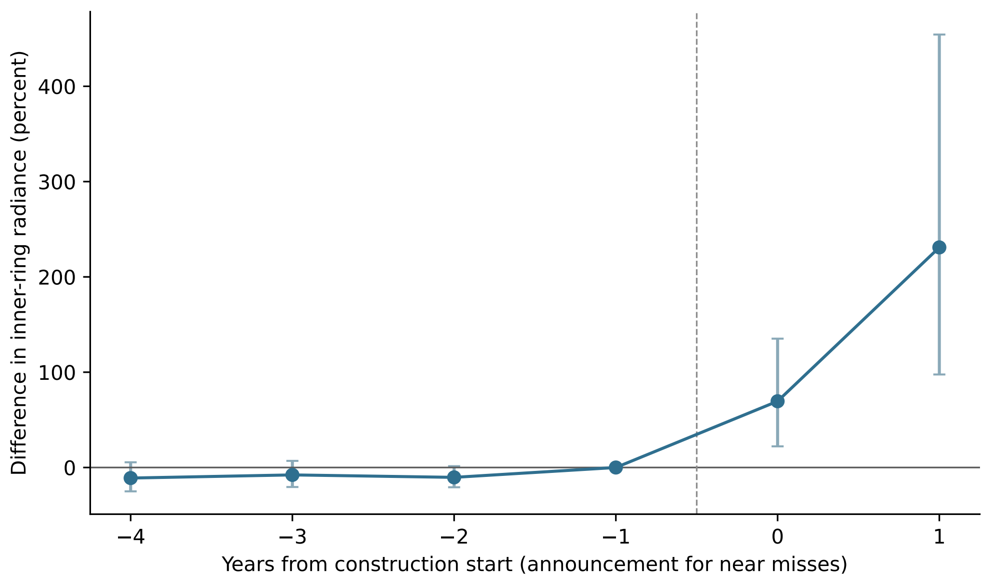

Figure 1: Capital footprint at built versus announced near-miss sites

Alt text: The event-study line is flat for four years before construction, rises sharply in the construction year, and rises further one year later. Confidence intervals widen after construction.

Notes: Event study of equation (1): the inner-ring (0–1 km) log nighttime-lights path at forty-three isolated built hyperscale facilities relative to forty-nine isolated announced near-miss facilities, omitting the year before. Built facilities are dated by satellite-detected land clearing; near-miss facilities are dated by announcement year. Standard errors are clustered on two-kilometer campus groups. Near-miss coverage is 48 sites in the construction year and 21 one year later. Table 2 reports observable differences between the groups.

4.2 Within-site ring design

The within-site ring design addresses two limitations of the built-versus-near-miss comparison. The first is dating. A hyperscale campus takes two to three years to build, so dating the event to the operational opening places the construction phase, when the parcel is physically changing, before the event window and reads it as a pre-trend. We date each facility instead to the construction-start year, recovered from satellite-detected land clearing (Section 2.1). Observing the construction start and the operational opening separately also prevents the construction period from being read as an opening-date pre-trend.

The second issue is the sample. Roughly two in five hyperscale sites sit within two kilometers of an earlier data center, such that the site inherits its neighbor's brightness in its inner ring before it breaks ground. Splitting the sample makes this explicit. At sites with an earlier neighbor within two kilometers, inner-ring brightness rises about fifty percent in the years before construction. At isolated sites, where the inner ring is genuinely just the future parcel, the pre-construction excess is seven percent, and the event study is flat before groundbreaking. A joint test of the pre-construction coefficients does not reject ($p = 0.21$), and the jump is thirty-eight percent in the construction year, surviving a control for a site-specific linear trend estimated from the pre-construction years.

The two designs agree on timing and shape. Both show a flat pre-period at isolated sites followed by a sharp jump in the construction year and a multi-year build-up. The point magnitudes are not directly comparable because the two designs transform radiance differently, but each implies a construction-year brightening of several tens of percent at the parcel.

4.3 Spatial decay

The footprint is sharply localized. Estimated ring by ring, the effect is largest at zero to one kilometer, falls by more than half at one to two kilometers, and is small and fading by two to five kilometers, with no detectable effect beyond five kilometers. A specification that omits the zero-to-one kilometer ring and uses one-to-two kilometers as the inner comparison returns the same decay, so the result is not an artifact of the campus site itself. Overall, the site impact is large but it does not reach far.

4.4 Colocation facilities as a comparison

Colocation facilities in our sample are generally smaller, multi-tenant projects and often occupy or expand existing facilities, although some are purpose-built. They show no comparable change in the capital footprint. The remainder of the paper therefore focuses on hyperscale facilities.

5 Test of agglomeration

Section 4 establishes that a hyperscale data center is a large, spatially concentrated investment. The question is whether that capital generates the local spillovers that large plant openings have historically produced. Greenstone et al. [2010] trace those spillovers to three channels: a deeper shared labor pool, denser supplier linkages, and knowledge spillovers among nearby firms. We test each channel at the facility level, using the within-site ring design for outcomes that pass its pre-trend test and the built-versus-near-miss comparison for job postings, which do not. Colocation facilities provide a second comparison, with the caveat that they sit in denser markets than hyper-scale campuses. We find no detectable opening-date response through these channels, although the hiring and supplier estimates are not precise enough to rule out moderate effects.

Figure 2 puts the preferred estimates on a common scale. The salary and firm-entry estimates are relatively precise. The hiring and supplier estimates are not: their intervals admit increases of roughly thirty percent, and their 80-percent minimum detectable effects are between 37 and 41 percent. Appendix Table 9 reports the corresponding intervals and minimum detectable effects.

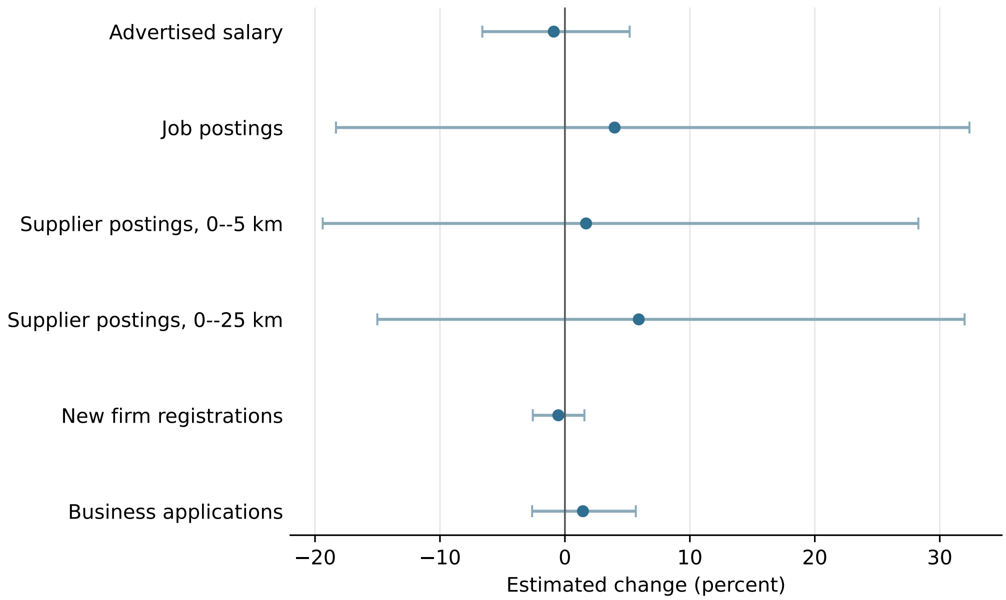

Figure 2: Precision of the agglomeration-channel estimates

Alt text: Horizontal confidence intervals for six estimates. Salary and firm-entry intervals are narrow around zero; job-posting and supplier intervals are much wider and include sizable positive effects.

Notes: Preferred estimate for each outcome with 95% confidence intervals, transformed from log points to percent. Job postings compare built sites with announced near misses. Salary estimates use within-site rings with a flat pre-period. Supplier and firm-entry estimates control for an inner-ring trend. The business-application estimate is the national county trend-break specification.

5.1 Labor pooling

If a data center deepens the local labor pool, hiring and wages should rise in the surrounding area. We use the built-versus-near-miss design for job postings because the within-site series carries a differential inner-versus-outer trend. Relative to near-miss sites, postings near built facilities move by +4.0% ($p = 0.75$) with a flat pre-trend ($p = 0.96$). The estimate remains statistically indistinguishable from zero using Poisson counts, a long difference that controls for pre-period volume, and construction dating. This is not a precise zero. Given the sample size and variance, the design could detect only a hiring increase of roughly forty-one percent or more, so it cannot rule out moderate gains.

Advertised salaries provide a narrower test. The mean log salary of nearby postings moves by -0.9% at 0–5 km ($p = 0.77$), with a ninety-five percent interval of $[-6.6, +5.2]$ percent. The panel is concentrated in salaried, posted positions, so the result is most relevant for white-collar workers and may not capture construction-trade compensation. The salary-disclosure rate is itself unchanged at opening ($p = 0.43$), so differential disclosure does not generate the result. Workplace employment from the Longitudinal Employer-Household Dynamics data carries a different limitation. Its raw within-site effect is $+8\%$ at 0–5 km, but a strong pre-existing trend explains the increase; detrending reduces the opening-date jump to -0.3%.

5.2 Supplier linkages

The second channel is local supplier linkages. We isolate postings in the sectors that build and equip a data center: construction and specialty trades, electrical-equipment wholesale, architectural and engineering services, and machinery manufacturing. The hyperscale supplier composite moves by $+1.7\%$ at 0–5 km ($p = 0.89$). Because supplier linkages may operate at the metropolitan level, we also compare the pooled 0–25 km zone to a 50–150 km control. That estimate is $+5.9\%$ ($p = 0.61$). The 95-percent intervals are $[-19, +28]$ percent at 0–5 kilometers and $[-15, +32]$ percent at the metropolitan scale. The results therefore do not establish a tight zero. They show that no large, precisely measured supplier cluster appears in the postings data.

5.3 Knowledge spillovers and firm entry

The third channel is knowledge spillovers among co-located firms. Appendix A maps heavy users of data-center capacity. They sit somewhat closer to data centers than the average firm, but much of this proximity reflects shared metropolitan location, and 69 percent of nearby compute firms pre-date the nearest facility. Exurban data centers often sit in places with few compute firms. These descriptive patterns do not identify an opening effect, but they give little indication of a cluster forming around the site.

New firm formation also shows no average jump at opening. We use secretary-of-state new-registration records for Virginia and Texas geocoded to the facility rings. The within-site jump at 0–5 km is $+0.7\%$ ($p = 0.24$) in the pooled sample and -0.5% ($p = 0.62$) for hyperscale facilities. A national county design using Census Business Formation Statistics also finds no effect ($+1.4\%$, $p = 0.49$). The upper bounds of the corresponding 95-percent intervals are 1.6 and 5.7 percent. We therefore find no systematic entry response. This does not imply that entry is zero in every location.

5.4 Incumbent productivity

In Greenstone et al. [2010], incumbent establishments experience total factor productivity gains of around twelve percent. We do not observe plant-level productivity. We instead ask whether advertised salaries or hiring at incumbent firms rise, two outcomes through which a large local productivity gain could appear. The advertised-salary interval rules out a wage response as large as the twelve-percent benchmark, provided that some of the productivity gain reaches posted salaries. An event study of hiring by firms that existed before the facility shows no positive response, but the pooled and colocation samples carry a pre-existing downward trend that the design cannot fully remove. We therefore read incumbent hiring as corroborating rather than causal evidence and do not claim to observe productivity directly.

5.5 What county-industry data can establish

County-industry data are a natural way to look beyond job postings, but the detailed Quarterly Census of Employment and Wages cells have two limitations in this setting. First, the Bureau of Labor Statistics suppresses employment when publication could reveal an individual employer. The public files record these cells as zero even though the employment level is not observed. Retaining the disclosure flag shows that suppression affects 68 percent of reported county-year records in data processing, 51 percent in telecommunications, and 66 percent in web and information services. Treating those values as observed zeros mechanically creates employment growth when a cell first becomes publishable.

Second, the host counties were already changing before the facility opened. We estimate cohort-by-event-time effects relative to the year before opening, using not-yet-treated counties as controls and a state-cluster bootstrap. Table 4 reports the average effect over event years zero through five and a joint test of the pre-period leads. Four of the ten hyperscale outcome series show statistically significant pre-opening changes at the five-percent level. These failures are concentrated in telecommunications, web and information services, and professional services. The data-processing establishment series is different: its pre-trend test does not reject, and the establishment count rises when the facility enters the county.

Table 4: County-industry disclosure and pre-trend analysis: hyperscale facilities

Industry (NAICS)	Employment suppressed	Employment effect	Pre-trend p	Establishments effect	Pre-trend p
Data processing (518)	68%	+5.2%	0.58	+27.2%***	0.11
Telecommunications (517)	51%	+5.2%	0.28	−2.6%	0.04
Web and information services (519)	66%	+9.3%	0.39	+18.3%***	0.03
Computer systems design (5415)	40%	−0.9%	0.85	+5.7%	0.11
Professional and technical services (541)	27%	+6.2%**	< 0.01	+7.6%***	< 0.01

Notes: Employment suppression is the share of reported county-sector-year records carrying the BLS non-disclosure code. Effects are cohort-weighted mean log changes over event years zero through five, transformed to percent, relative to not-yet-treated counties. Pre-trend p is a joint test for event years -4 through -2. Inference uses 1,999 state-cluster bootstrap draws. The 519 series ends in 2021 because the 2022 NAICS revision moved major components to sectors 513 and 516. Stars refer to the post effect: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Benjamini–Hochberg q-values across the ten sector-outcome tests are < 0.001 for data-processing establishments, 0.011 for web and information-services establishments, 0.066 for professional-services employment, and < 0.001 for professional-services establishments.

The county data identify one direct margin. Data-processing establishments rise by 27 percent, and the result survives a Benjamini–Hochberg adjustment across the ten sector-outcome tests. This is the sector that contains the facility itself, so the increase confirms direct entry rather than downstream agglomeration. Data-processing and telecommunications employment do not show a precise increase once we preserve the suppression flags. The positive establishment estimates in web and information services and in professional services also survive the multiple-testing adjustment, but both have significant pre-period leads. We therefore do not use them as evidence of downstream growth. The county data distinguish direct entry from broader industry effects, but do not identify a downstream cluster.

6 Incidence outside the labor market

The housing evidence is suggestive rather than definitive. The within-site rent estimates are +6.6% at 0–5 km, +9.4% at 5–10 km, and +8.5% at 10–25 km. A density-matched placebo shows that comparable non-data-center locations also experienced substantial rent growth during the 2019–2023 work-from-home period, so the within-site estimates are upper bounds. Facility-level synthetic controls return smaller point estimates, between two and five percent across the three rings, although we cannot use this procedure for formal placebo inference. The built-versus-near-miss estimate is +4.3% ($p = 0.23$), with a flat pre-trend ($p = 0.44$). We therefore read the evidence as consistent with a modest rent response, with considerable uncertainty about its size. Colocation rent estimates show no detectable effect.

Neither housing supply nor net migration rises. A place-level Census Building Permits Survey analysis finds no multifamily permitting response, with permits for five or more units up 3.4% ($p = 0.87$). For migration, we assign each county its first reliably dated hyperscale opening within 25 km, including openings before the IRS panel, and estimate cohort-by-event-time effects using not-yet-treated counties. Gross inflows rise by 5.4% and gross outflows by 7.6%, but the outflow series has a significant pre-trend. In-migrant income also rises before opening. We therefore do not interpret the gross-flow estimates as opening effects. The log inflow-to-outflow ratio for returns moves by -1.4% over event years zero through five ($p = 0.52$), with a joint pre-trend p-value of 0.14. The analogous adjusted-gross-income ratio moves by $+0.4\%$ ($p = 0.91$), with pre-trend $p = 0.09$. These estimates provide no evidence of net in-migration. The pre-opening trends prevent a causal interpretation of the gross migration or migrant-income estimates.

Foot traffic and commercial spending also show no robust change relative to near-miss sites. The foot-traffic estimate is about -12% ($p = 0.12$) on the full panel but falls to -3% ($p = 0.68$) on the balanced panel. Commercial spending moves by $+4.6\%$ ($p = 0.60$). The within-site spending estimate is positive and passes a random-donor comparison, but it is imprecise and the built-versus-near-miss design does not confirm it.

The fiscal estimates are concentrated in school finance rather than broad local-government spending. Table 5 reports the same not-yet-treated estimator and state-cluster bootstrap used in the county analyses. Local school property-tax revenue rises by 16.6% ($p = 0.07$) with no detectable pre-trend. Capital outlay per pupil rises by 28.5%, but is imprecisely estimated ($p = 0.18$). Total local revenue and local revenue per pupil do not rise. The county highway estimate is positive and marginally significant, but moves materially when the opening dates are updated. The broad public-works series fails its pre-trend test. Neither the property-tax nor highway estimate survives adjustment across the six fiscal outcomes ($q = 0.21$ for each). We therefore treat the fiscal results as suggestive rather than as robust effects.

Table 5: Public-finance effects after hyperscale openings

Outcome	Geography	Effect	Post p	FDR q	Pre-trend p
Local property-tax revenue	school district	+16.6%	0.07	0.21	0.31
Total local revenue	school district	−2.3%	0.83	0.83	0.55
Local revenue per pupil	school district	+2.0%	0.76	0.83	0.14
Capital outlay per pupil	school district	+28.5%	0.18	0.35	0.57
Highway current + construction	county, balanced	+38.9%	0.07	0.21	0.15
Broad public works	county, balanced	+9.3%	0.40	0.61	< 0.01

Notes: Cohort-weighted mean log changes over event years zero through five, transformed to percent, relative to not-yet-treated geographies. School data are NCES/Census F-33 finance files for 2012–2023; county data are Census local government finance files for 2012–2023. Outcomes are adjusted for state-by-year means before the cohort comparisons. Inference uses 1,999 state-cluster bootstrap draws. Pre-trend p jointly tests event years -4 through -2. FDR q applies the Benjamini–Hochberg adjustment across the six outcomes in the table. The highway estimate moves materially across alternative opening dates and is treated as a sensitivity result rather than a headline effect.

Taken together, the non-labor outcomes do not point to a broad local demand boom. The rent point estimates are positive and the fiscal point estimates are suggestive, but we find no net migration, permitting, foot-traffic, or spending response that would indicate a large expansion of the local economy.

7 Interpretation

The results separate the physical scale of the investment from its propagation to nearby workers and firms. The capital footprint is large within the first kilometer and fades rapidly with distance. Data-processing establishments rise when the facility enters the county, which records the direct facility. Advertised salaries and new-firm formation show no opening-date increase, and these estimates are precise. Hiring and supplier postings are also centered near zero, but their confidence intervals admit moderate effects. The evidence therefore rules out a broad firm-entry or advertised-wage response more clearly than it rules out hiring or supplier gains.

The pattern is consistent with the technology of a data center. A hyperscale campus employs dozens of workers rather than thousands. Its main inputs include electricity, network connectivity, construction services, and specialized equipment, much of which can be procured from regional or national suppliers. Its output is delivered to distant customers over the network. These features weaken the local labor-pooling, supplier, and customer-linkage channels that operate around a large manufacturing plant. They also help explain the geography of compute. Data centers and compute firms share metropolitan infrastructure and skilled labor, but 69 percent of compute firms within 25 kilometers predate the nearest facility. The co-location is therefore not, by itself, evidence that the facility created the cluster.

The non-labor evidence is more suggestive. Rent point estimates are positive in the synthetic comparisons, while net migration and multifamily permitting do not rise. School property-tax revenue and highway spending also have positive point estimates, but neither survives adjustment across the fiscal outcomes. Residential electricity rates do not rise for the cloud-era cohorts in our sample. These outcomes identify possible incidence margins, not a welfare accounting. We do not observe land prices, environmental damages, or the terms of utility and tax agreements.

Hyperscale and colocation facilities differ in both technology and location. Hyperscale campuses are generally larger and more likely to involve greenfield construction, while colocation facilities are multi-tenant and concentrated in denser markets. Their different outcome paths are informative, but they combine facility-type and market differences and should not be read as if facility type were randomly assigned.

8 Conclusion

A hyperscale data center rivals the manufacturing plants in the agglomeration literature in capital scale, but its visible local response is narrower. At isolated sites, brightness rises when construction begins and fades within five kilometers. Announced near misses show the same timing, although they are not balanced runner-up sites. Daytime imagery confirms that physical change begins with construction.

The labor and firm evidence separates direct entry from downstream clustering. Data-processing establishments rise with entry. Advertised salaries and new-firm registrations do not, with upper confidence bounds of 5.2 and 1.6 percent. Hiring and supplier estimates are also near zero but less precise. Compute firms locate disproportionately close to data centers, yet most arrived first and exurban campuses often have few such firms nearby. The shared geography reflects existing clusters more clearly than new clustering around the facility.

The non-labor outcomes do not imply a broad local demand boom. Rent point estimates are positive, but multifamily permitting and net migration do not rise. Fiscal estimates do not survive adjustment across outcomes, and foot traffic, commercial spending, and residential electricity rates show no robust increase. A hyperscale project clearly brings large physical investment. For the cloud-era cohorts we study, we do not detect the nearby worker and firm responses needed to support a broad local-multiplier claim. Larger post-2022 artificial-intelligence campuses fall outside our sample.

A The geography of compute

A common rationale behind data center incentives is that the facility will anchor a local digital economy: firms that use cloud capacity will locate nearby. We examine this claim by mapping where heavy users of compute locate relative to data centers and asking whether the shared location predates the facility. We use the Veridion firmographic database [Veridion, 2026], which geolocates 6.5 million US establishments and classifies each by industry, employment, and founding year. We define compute users by primary industry. Software publishing, computing infrastructure and hosting, computer systems design, and web and streaming services form the compute-native core, with finance, scientific R&D, and semiconductors as a broader set.

A.1 Compute users cluster near data centers

Compute users are closer to data centers than the average firm. The median compute-native firm sits 11 kilometers from the nearest data center, against 20 kilometers for non-compute firms, and 69 percent of compute users fall within 25 kilometers of a data center, against 55 percent of other firms (Table 6). The association varies by industry. Cloud and hosting firms are nearest at a 6 kilometer median, software firms next at 9 kilometers, while telecommunications and streaming, whose footprints are more dispersed, sit farthest out.

This proximity is not an artifact of headquarters location. Measuring each firm's operating footprint by the geography of its job postings rather than its headquarters returns the same pattern at a slightly larger distance, because a firm's branch offices spread beyond its headquarters metro. Nor is it merely that both compute users and data centers crowd into a handful of tech metros. Removing the 45 largest metropolitan areas, compute users remain closer to the surviving data centers than non-compute firms are (a 75 versus 87 kilometer median, with software firms among the closest), so the association holds in second-tier and rural America as well.

Table 6: Distance from firms to the nearest data center

	Firms	Median km	≤ 25 km	≤ 50 km
Panel A: Headquarters location (Veridion)				
Non-compute firms	5,692,641	20.5	55%	70%
Compute users (core)	309,760	11.0	69%	81%
Cloud and hosting	26,188	6.1	75%	84%
Software	101,757	8.9	72%	84%
Computer systems design	154,934	11.7	68%	81%
Finance	193,540	15.9	62%	75%
Telecommunications	10,125	22.2	53%	66%
Panel B: Establishment footprint (job postings)				
Compute users (core)		15.1	63%	76%
Panel C: Outside the 45 largest metros				
Non-compute firms	2,603,195	86.8	20%	33%
Compute users (core)	103,356	75.3	30%	41%

Notes: Great-circle distance from each firm to the nearest reliably dated hyperscale data center. Panel A places each firm at its Veridion headquarters coordinate. Panel B replaces the headquarters with the firm's job-posting locations (WageScape), weighted by the number of distinct hiring firms at each location, so it reflects the operating footprint rather than the headquarters. Panel C excludes all firms and data centers within 50 km of the 45 largest metropolitan areas and technology hubs and measures distance to the nearest remaining data center. Compute-native "core" sectors are software publishing, computing infrastructure and hosting, computer systems design, and web and streaming services.

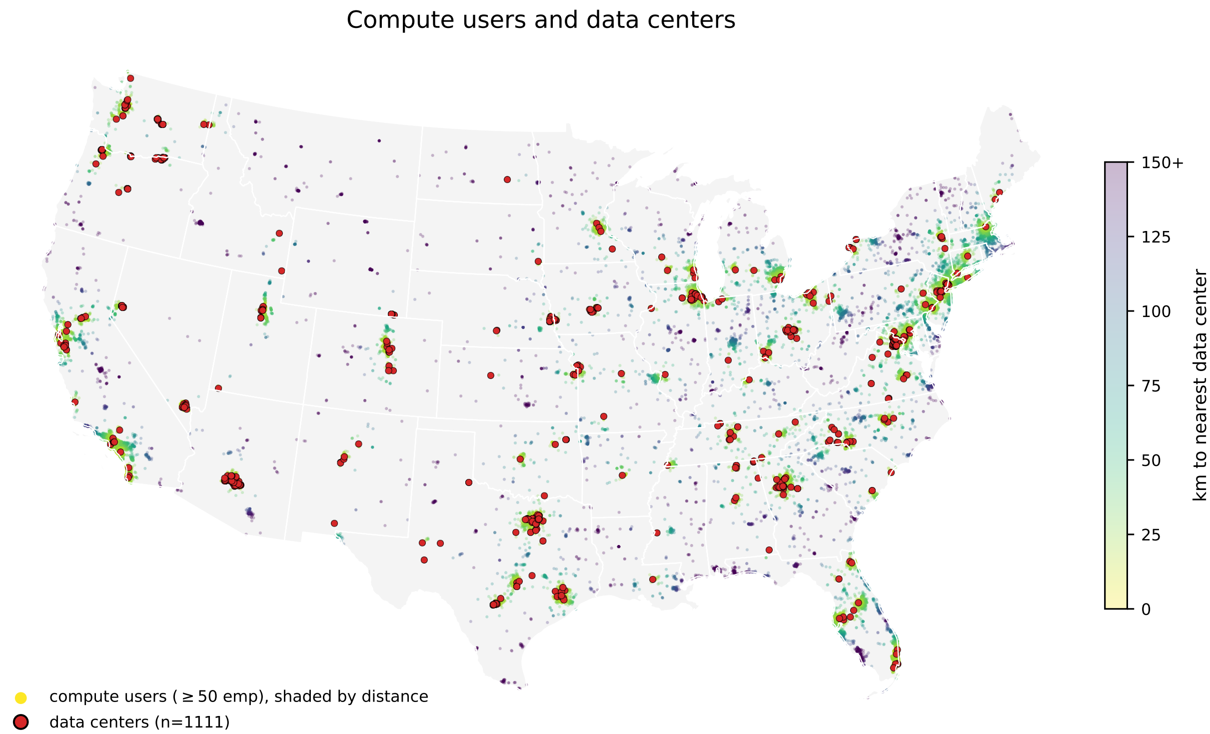

Figure 3: Compute-user firms and data centers across the continental United States. Points are compute-native firms with at least 50 employees, shaded by distance to the nearest data center, with data centers marked in red. Both concentrate in the same coastal and metropolitan corridors. Points in the interior are darker because they are farther from a facility.

Alt text: A US map shows compute firms and data centers concentrated in coastal and metropolitan corridors, with sparse coverage across much of the interior.

The two populations trace the same map (Figure 3), crowding into the coastal and metropolitan corridors and thinning together across the interior. The association is strongest for the largest firms. When sorted by employment, the median distance to a data center falls monotonically from 13 kilometers for the smallest firms to 4 kilometers for those with more than ten thousand employees, and the spread narrows with it. The biggest compute users cluster tightly around data centers, and almost none are far from one. Only about 5 percent of compute firms with at least a thousand employees sit more than 50 kilometers from any data center. The handful that do are software companies that scaled in place in second-tier cities, such as Esri in Redlands, Jack Henry in rural Missouri, and Ansys outside Pittsburgh. These examples show that large compute users need not locate near a data center to access cloud capacity.

A.2 Two kinds of data center hub

The geographic overlap is primarily a metropolitan phenomenon. Grouping data centers into hubs reveals a sharp split. Metropolitan hubs are embedded in dense clusters of sector-matched compute users. The 239 data centers around Ashburn, Virginia sit among thousands of government IT contractors, with DXC, Leidos, Booz Allen, SAIC, and CACI all within 25 kilometers. The Bay Area hubs are ringed by Google, Broadcom, Synopsys, and ServiceNow. By contrast, the large exurban hyperscale clusters are isolated. The 50 data centers around Hillsboro, Oregon and the cluster near Quincy, Washington each have on the order of four compute firms within 25 kilometers. The same type of physical infrastructure coincides with a dense compute neighborhood in a technology center and a sparse one in the exurbs.

A.3 Shared location does not imply data-center-led clustering

Part of the explanation is shared infrastructure. Data centers and compute users both require access to the high-voltage grid, and both sit closer to it than ordinary firms. The median data center is 3.4 kilometers from a 230-kilovolt substation, the median compute user 5.0 kilometers, and the median other firm 6.4 kilometers. High-voltage substation density independently predicts where compute users locate. But the grid is only part of the story. Compute users remain closer to data centers after controlling for substation access. A shared pull toward the power backbone, fiber, and skilled labor can therefore place both in the same markets without one drawing the other.

Three patterns weigh against the interpretation that a data center creates the local compute cluster. First, the compute users near data centers predate them. Among compute firms within 25 kilometers of a data center, at least 69 percent were founded before the nearest data center opened. Because the firm registry over-represents recently founded firms, the share is likely higher. The local compute base usually existed first. Ashburn's history is consistent with this ordering: Northern Virginia's government technology firms and early internet exchange preceded the region's data-center cluster. Second, the job-posting analysis finds no detectable increase around opening, either in compute-using industries or overall, although the estimates are imprecise. Third, exurban data centers are located in areas with few compute firms. The median metropolitan data center has roughly 2,700 compute firms within 25 kilometers, while the median exurban data center has 41 (Table 7).

Table 7: Compute users near metropolitan versus exurban data centers

	Data centers	Median compute firms ≤ 25 km	Share with < 5 nearby
Metropolitan data centers	824	2,702	0%
Exurban data centers	287	41	11%

Notes: Each data center is classified as metropolitan if it lies within 50 km of one of the 45 largest metropolitan or technology-hub centers, and exurban otherwise. The middle column reports the median count of compute-native firms (Veridion) within 25 km of the data center; the last column reports the share of data centers with fewer than five compute firms within 25 km. Separately, among compute firms within 25 km of any data center, 69 percent were founded before the nearest data center opened.

A.4 Implication for place-based policy

The evidence does not support the claim that a data center, especially an exurban one, will draw compute users to the surrounding area. Firms that consume cloud capacity locate near transmission, fiber, and skilled labor. These assets concentrate in metropolitan areas, and a single data center does not create them. The compute users already near data centers were mostly there first, and we find no systematic increase in firm entry after opening. The clearest measurable local response is the capital footprint, with more modest evidence of higher rents; we do not detect a cluster of digital firms.

B The instrument search

A natural way to address the siting-selection problem is an instrument for data center location. We searched for one and did not find a credible one. This appendix documents the search because the failure is informative. It is the reason the within-site design carries the main spatial interpretation, while announced near misses provide a benchmark rather than an instrument. The exercise also shows the limits of common siting predictors.

Predetermined siting instruments. We tested several plausibly exogenous, predetermined predictors of data-center siting, interacted where needed with the national capacity wave. We also applied a manufacturing falsification test: a valid instrument for data-center activity should not independently predict manufacturing employment. Table 8 summarizes the results. Transmission-substation density predicts siting strongly but fails this test because it is a proxy for industrial land. The 1980 urban college share, which works in the all-data-center county literature, carries the wrong sign for hyperscale, because hyperscale campuses go to low-college exurban land rather than to educated metros. Long-haul fiber proximity predicts siting only across metropolitan areas, not within a county, so it cannot identify a sub-county effect. Climate, measured by wet-bulb temperature and free-cooling days, does not predict siting at all, because the largest US hyperscale hub is hot and humid and operators engineer around climate to chase cheap power. Generation-capacity timing has no within-county predictive power. The features that predict hyperscale siting (power, industrial land, fiber) are the same features that are correlated with local industrial character. The features less likely to violate the exclusion restriction (climate and historical human capital) do not predict where these facilities go. We therefore do not find a credible predetermined instrument for hyperscale siting at the sub-county scale.

Table 8: Predetermined siting instruments and why each fails

Instrument	First stage	Why it fails
345 kV substation density	strong	predicts manufacturing employment
1980 urban college share	moderate	predicts fewer hyperscale sites
InterTubes fiber proximity	metro-scale	no within-county effect
Climate (wet-bulb, free-cooling)	none	does not predict siting
Generation-capacity timing	none	does not predict siting within counties

Notes: Each instrument is the predetermined county feature interacted with the national hyperscale capacity wave. Predicting manufacturing employment violates the exclusion restriction.

A shift-share instrument also fails. We also construct a shift-share instrument of the form used in concurrent county-level work [Alvarez et al., 2026], interacting predetermined fiber and college exposure with national data-center demand growth. This approach does not solve the problem because our public measures of data-center scale—capacity, square footage, and facility count—are lumpy and supply-driven. The first-stage F-statistic with square footage as the endogenous variable ranges from twenty to thirty, but the exclusion concerns described above remain.

Because the shift-share design does not resolve the exclusion concerns, we do not use its estimates as evidence. The instrument also predicts population-serving sectors such as health, reinforcing the concern that it captures general area development. We therefore report the exercise as documentation of the search rather than as an identification strategy.

C Incidence on electricity rates

A central concern in data-center siting debates is that a hyperscale facility's load will raise electricity bills for nearby households. Existing estimates do not point in one direction. Wholesale-market modeling finds higher wholesale prices [Taylor et al., 2026], and county-level instrumental-variable estimates find higher local electricity prices [Alvarez et al., 2026]. In contrast, Watten et al. [2026] estimate that data-center growth modestly reduced average retail rates from 2015 through 2024. Whether residential customers in a host utility's territory pay more is therefore an empirical question. Data centers select utilities with cheap power and available capacity, so a comparison of treated and untreated utilities must account for the siting choice.

We treat each utility within each state as a separate unit. Using Energy Information Administration Form 861, we compute average residential prices as revenue over megawatt-hours sold from 2012 through 2024. A unit is treated in the year the first hyperscale facility opens in a county it serves. The hyperscale sample contains 52 utility-states first treated between 2015 and 2021 and no colocation facility. We compare them with roughly 2,000 utility-states that host no facility in the registry and, separately, with 54 utilities serving counties with an announced hyperscale near miss.

Figure 4 reports a cohort-by-event-time estimator using not-yet-treated controls. Against never-treated utilities, the mean effect over years zero through five is -1.0% ($p = 0.42$), with a joint pre-trend p-value of 0.97. The balanced panel, which retains utility-states observed in at least twelve of the thirteen years, gives -1.5% ($p = 0.17$). Against near-miss utilities, the full sample estimate is -0.9% ($p = 0.37$), and the year-five estimate is -1.9% ($p = 0.27$). Colocation estimates are small and positive, with a mean effect of $+0.7\%$ ($p = 0.54$).

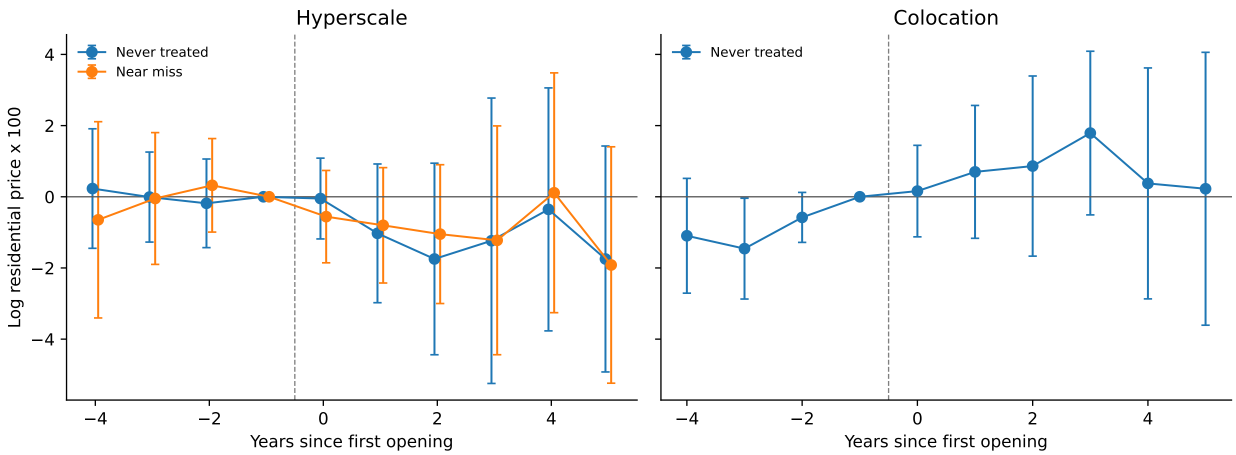

Figure 4: Residential electricity prices around the first data-center opening in a utility's territory

Alt text: Two event-study panels show residential-price estimates around hyperscale and colocation openings. Hyperscale estimates become modestly negative after opening; colocation estimates fluctuate around zero with wide intervals.

Notes: Cohort-by-event-time estimates of log average residential price (revenue over MWh, EIA Form 861 bundled-service utilities, 2012–2024). Hyperscale: 52 treated utility-states, 2,040 never-treated controls, and 54 near-miss controls. Colocation: 60 treated utility-states and the never-treated controls. The never-treated comparisons remove state-by-year means; the near-miss comparison removes year means. Inference uses 1,999 state-cluster bootstrap draws. Reference period is the year before opening; 95% confidence intervals shown.

The estimates provide no evidence that host-utility residential rates rise over this horizon. The negative point estimates are consistent with fixed network costs being spread over a large new load, but the full-sample and near-miss averages are imprecise. We do not observe the utility's cost allocation or contract, so the design does not identify that mechanism. Retail rates are also only one incidence margin: electricity generation creates environmental costs that may be borne outside the host utility [Muller, 2026]. The results cover cloud-era cohorts through 2021 followed to 2024, not the larger post-2022 artificial-intelligence build-out.

D Supplementary results

Table 9 reports confidence intervals and minimum detectable effects for the preferred agglomeration-channel estimates.

Table 9: Precision of the agglomeration-channel estimates

Outcome	Estimate	95% confidence interval	80% MDE
Advertised salary	-0.9%	[-6.6, +5.2]	8.9%
Job postings	+4.0%	[-18.3, +32.4]	41.2%
Supplier postings, 0–5 km	+1.7%	[-19.4, +28.3]	39.4%
Supplier postings, 0–25 km	+5.9%	[-15.0, +32.0]	37.0%
New firm registrations	-0.5%	[-2.6, +1.6]	3.0%
Business applications	+1.4%	[-2.6, +5.7]	6.0%

Notes: Preferred estimate for each outcome, transformed from log points to percent. MDE is the minimum two-sided effect detectable with 80% power at the five-percent level, holding the reported standard error fixed. Job postings compare built sites with announced near misses. Salary estimates use within-site rings with a flat pre-period. Supplier and firm-entry estimates control for an inner-ring trend. The business-application estimate is the national county trend-break specification.

Data Availability Statement

This study combines publicly available data with several proprietary datasets obtained under license. All code that constructs the analysis datasets from the raw sources and reproduces every table and figure will be deposited in a public data and code repository. The deposit includes the publicly available inputs and all author-constructed files, namely the data-center registry, the operating and construction-start dates, and the announced near-miss cohort. The proprietary inputs cannot be redistributed. For each, we report the provider and access terms and include the exact product identifiers, date ranges, geographic filters, and extraction code, so that a licensed user can regenerate the facility-by-ring panels and reproduce all results.

Publicly available data.

- Satellite nighttime lights: VIIRS Day/Night Band Monthly Composite (NOAA, product VCMCFG), accessed through Google Earth Engine.
- Daytime imagery for construction dating: Sentinel-2 and Landsat surface reflectance (ESA and USGS), accessed through Google Earth Engine.
- Data-center registry: starting point from the IM3 Open Source Data Center Atlas (Pacific Northwest National Laboratory), augmented by the authors from operator press releases, state economic-development announcements, the Good Jobs First subsidy tracker, trade press, DatacenterMap, Cloudscene, and data-center REIT filings. The compiled registry, dates, and announced near-miss cohort are in the deposit.
- Construction timing: Federal Aviation Administration Obstruction Evaluation / Airport Airspace Analysis filings (Form 7460-1).

- Employment and establishments: Quarterly Census of Employment and Wages (Bureau of Labor Statistics) and County Business Patterns (Census Bureau).
- Migration and income: IRS Statistics of Income county-to-county migration data.
- Housing supply: Census Building Permits Survey. Population: 2020 Census Centers of Population (tract population-weighted centroids).
- Workplace employment: LEHD Origin-Destination Employment Statistics (LODES, Census Bureau).
- Business formation: Census Business Formation Statistics, and state business-registration bulk files for Virginia, Texas, and Iowa (Secretaries of State and data.iowa.gov).
- Local public finance: Census of Governments and the Annual Survey of School System Finances (F-33).
- Electricity: U.S. Energy Information Administration Form 861.
- Instrument-search inputs: the InterTubes long-haul fiber map [Durairajan et al., 2015]; the 1980 county urban-college share (IPUMS NHGIS); and transmission-substation locations (Homeland Infrastructure Foundation-Level Data).

Proprietary data (available under license, not redistributable).

- Job postings: LinkUp Job Records [LinkUp, 2025]; WageScape salary postings [WageScape, 2022].
- Residential rents: RentHub Rental Data [RentHub, 2022].
- Consumer spending: SafeGraph Spend Patterns [SafeGraph, 2022].
- Foot traffic: Advan Weekly Patterns [Advan Research, 2025].
- Firmographics: Veridion firm master data.

The five Dewey-distributed datasets are available to researchers through a subscription to Dewey Data (`https://www.deweydata.io`); Veridion firmographics are available under license from Veridion (`https://veridion.com`). The authors accessed these data under institutional license and are not permitted to redistribute the raw files. The deposit provides the dataset identifiers and the extraction and aggregation code required to rebuild the analysis files from a licensed copy.

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